

Risk policy memo — aluminium scrap import desk (SIM)

To: Risk Committee (SIM) · **From:** Head of Desk (SIM), Meridian Non-Ferrous Trading (SIM) · **Date:** 7 November 2022 · **Re:** limits for the next book, set against what the Mar-Oct 2022 book actually did

ACADEMIC SIMULATION — not actual trades. Book P&L was ₹192.1 m at 31-Oct-2022, but it is **not sign-robust**: –₹105.4 m to +₹334.3 m across the registered domestic anchor-premium grid (break-even –38,702 ₹/t inside it). So every limit below is sized on the risk the book ran, not on what it earned. Evidence is from the published tables; checks that include the MCX leg skip the 16 MCX exit/roll position dates P3 flags. Hedge effectiveness is measured on a unit-beta MCX proxy, so the MCX basis is under-represented in every hedge number here. The third-party mirror's beta is 0.75 on weekly closes and 0.45 on daily data, which the MCX evening close biases down: mean GARCH VaR is ₹8.5 m at the weekly beta and ₹17.3 m at the daily one (the pessimistic end, not a hedge ratio anyone would use), against ₹4.27 m on the proxy.

Area	Limit / rule proposed	Book evidence (2022)	Book vs limit → remediation
Position	Gross physical LME exposure ≤ 5,000 MT and ≤ USD 15 m	Peak 4,450 MT / USD 14.5 m on 21-Apr; unsold cargo 4,980 MT on 6-Apr	Within by construction — the limit was set just above the book's own peak, so this is not evidence; not raised, because that size already used the whole working-capital line (Liquidity row)
Unhedged	Net LME delta ≤ 1,000 MT and ≤ USD 3 m; net ≤ 25 % of physical once physical ≥ 1,000 MT	Absolute net: 95th pct 967 MT; max 1,817 MT (USD 6.36 m) on 8-Mar, the naked session before T01's hedge	MT breached 5 days (8-Mar-6-Jul); USD breached 2 days (8-Mar-14-Apr); % breached 26 of 99 days (ratios drift between re-sizings) → rebalance within 2 sessions; no purchase before its hedge can be placed
Hedge	MCX ratio 0.80-1.10 of LME-equivalent physical at every decision; roll ≥ 7 trading days before expiry, roll gains never budgeted (proxy curve is always contango); FX forwards ≥ 90 % of each fixed USD payable, on physical + forwards only	8 tickets filled 0.899-1.002; T04 at 0.50; 13 of 13 rolls by the deadline; FX 100 % except T06 60 %	T04 and T06 breach → re-size, or a committee exception signed before the trade
Stop-loss	Ticket –₹10 m from its own peak: same-day review. Ticket –₹20 m: hedge to 1.0, no same-grade additions. Book –₹40 m from peak: no new purchase until the committee reviews	Review level hit by T08 –₹26.1 m, T05 –₹14.0 m, T07 –₹11.9 m, T04 –₹10.9 m; book –₹49.9 m (9-Jun→8-Aug), past –₹40 m from 1-Aug	T08 hit its hard stop on 1-Aug, the book its stop from 1-Aug; T09 bought 3-Aug would have been refused (same grade as T08). Honest cost: T09 finished +₹10.0 m on base marks. The levels were chosen with these drawdowns in view, so which tickets trip is illustration, not a test
VaR	1-day 95 % VaR ≤ ₹10 m on the larger of GARCH(1,1) and 250-day historical (docs/40: the historical is the floor, because GARCH decayed below it into the April-May crash). Covers LME, FX and freight only: grade spread (daily σ ₹2.2 m) and the cash-3M spread (memo VaR +18 %) sit outside it	120 position days: GARCH mean ₹4.27 m, max ₹43.5 m on 9-Mar	Breached once (9-Mar) on clean days (the naked session); 5 days counting roll-day artefacts → cut the driving position the same day
Counterparty	Band from the illustrative score: A max use 100 %, B 90 %, C 75 % + security + advances ≤ 0.5x limit, D line cut to 50 %, advance/CAD only; never lift a hedge before an advance lands	BUY_RJK_01 band D (model PD 41.8 %, synthetic-data model); advances > 1.0x its ₹120 m line on 37 days, peak 1.83x, contracted 2.57x; 5 of 10 bookings outside the bands	Breached → BUY_RJK_01 line ₹120 m → ₹60 m, advance or CAD only; BUY_JNPT_01 band C: freeze; BUY_MUN_01 band C: freeze
Freight	FOB freight fixed ≤ 10 trading days after purchase; unhedged with a stop ≤ fixture + 20 %; no proxy hedge (no instrument tracks the India lanes; the two lanes are one factor)	Fixed after T01 8, T04 6, T07 2 days; stops +18.4 %, +12.7 %, +14.2 %; freight VaR mean ₹0.05 m; hypothetical +40 % spike costs T01 –₹8.55 m	Within
Liquidity	Working-capital line ₹1,000 m and LC line ₹1,000 m (sized on the Feb-2022 plan). Funding need + max(₹150 m, 99 % 10-day margin-at-risk at 15 % IM) ≤ line, checked daily; no LC opened that takes the LC line over limit	Peak need ₹1,179.0 m on 20-Jul; LC peak ₹1,414.8 m on 13-Jun; margin-at-risk peak ₹304.4 m on 21-Apr; crash fortnight (22-Apr-9-May) actually paid in +₹65.0 m of VM, but LME +23 % against the MCX shorts calls ₹181.8 m more and takes headroom from ₹0.2 m to –₹181.6 m while the cargo gains ₹427 m of mark, not cash	Buffer rule breached 30 days (10-May-5-Aug); line breached 19 days (16-May-5-Aug); LC breached 17 days (28-Apr-21-Jun) → pre-agree ad-hoc limits of ₹500 m (WC) and ₹500 m (LC) for peak months, or stagger usance maturities and purchases

Decisions requested. (1) Adopt the limits above for the next book. (2) Approve the BUY_RJK_01 line cut and the advance-before-hedge-lift rule now. (3) Mandate the treasury to seek the ad-hoc lines before the next book's peak months; until they are in place, no ticket is signed that the daily liquidity rule shows breaching the line within its cash cycle.

What these limits cannot see. The position, unhedged and stop levels were set with the 2022 book in view, so where the book lands against them is illustration, not evidence; the VaR limit excludes grade spread, the book's largest bucket after deal margin. The MCX series and USD/INR are proxies; grade factors and freight levels are hindsight reconstructions; credit PDs come from a model fitted to synthetic data; no drawing-power (stock and receivables) test is modelled; the stop-loss rules would not have fixed the drawdowns driven by grade spread and carry, only stopped the desk adding to them.